

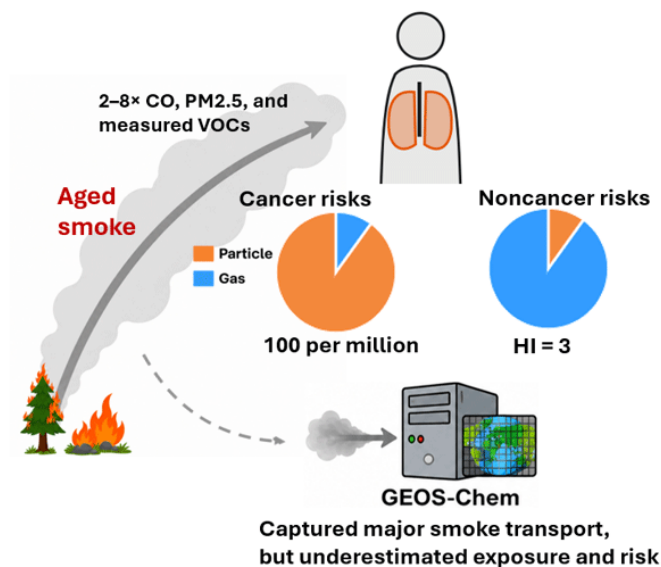
What Happens to Wildfire Smoke After Days in the Air?

Hourly ground measurements reveal what multi-day-old smoke brought to Missoula

Jin et al. (2026) | Atmospheric Chemistry and Physics

Large collaborative field campaigns have produced detailed measurements of wildfire smoke near fires and aboard research aircraft. Far less is known about what remains in smoke after it has traveled and reacted for several days before reaching people at ground level. In September 2020, Missoula, Montana, experienced three smoke events. During smoke-affected hours, fine-particle pollution ($\text{PM}_{2.5}$) averaged $43 \mu\text{g}/\text{m}^3$, about seven times the local background, and reached an hourly peak of $120 \mu\text{g}/\text{m}^3$. Much of the regional smoke had already spent several days in the atmosphere, although nearby fires may also have contributed. With hourly ground-level measurements, we asked two questions: what did the smoke contain when it reached the city, and how had its chemistry changed along the way?

Pollution remained far above the local background even after the smoke's chemical mixture had changed along the way.



At ground level, $\text{PM}_{2.5}$, carbon monoxide, and measured VOCs reached roughly two to eight times their local background levels.

Paper key visual from Jin et al. (2026), CC BY 4.0.

What we measured: Every hour, we measured $\text{PM}_{2.5}$, carbon monoxide, nitrogen oxides, ozone, and 75 volatile organic compounds (VOCs) at ground level. Satellite images and reconstructed air pathways helped us identify three smoke periods influenced by overlapping plumes. We then compared the measurements with GEOS-Chem, a widely used air-quality model.

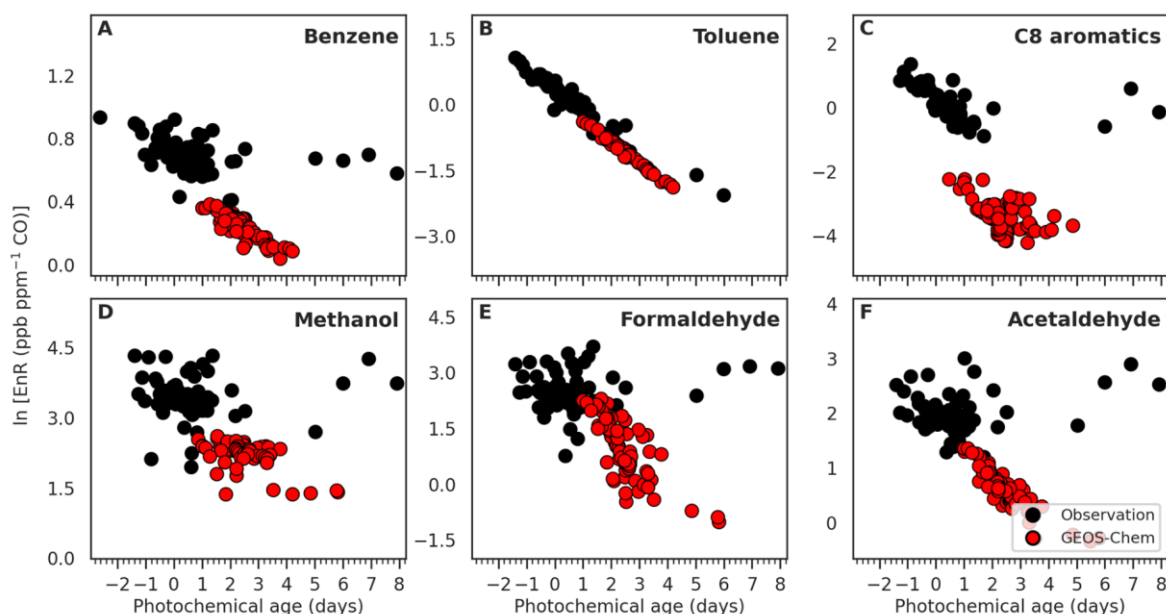
A CHEMICAL RECORD AT GROUND LEVEL

Ground Measurements Show How the Smoke's Chemistry Changed Along the Way

Even after days in the atmosphere, the smoke still carried a chemical record of what it was like near the fire and how it changed along the way.

To read that record, we compared multiple volatile organic compounds (VOCs) with carbon monoxide, a gas that changes relatively slowly. This helped us separate chemical change from simple dilution as the smoke mixed with cleaner air. We then used the disappearance of faster-reacting gases as a chemical clock. Chemical age tells us how far the smoke's chemistry had progressed, not exactly how long it had been traveling. Benzene and toluene declined predictably as the smoke aged. For these two gases, the near-fire signatures reconstructed from the ground data agreed within about 20% with aircraft measurements made close to western U.S. fires.

Other gases told a different story. Methanol, formaldehyde, and acetaldehyde declined more slowly. The flatter trends are consistent with continued formation partly offsetting chemical losses, although mixing with non-fire air may also have contributed. GEOS-Chem generally predicted too little of these VOCs relative to carbon monoxide. The model's OH exposure was about twice the value inferred from the observation-based chemical clock. Because OH removes many VOCs, the modeled smoke aged too quickly and some gases disappeared too fast. The comparison also suggests that the model missed some chemistry that formed oxygenated gases as the smoke traveled.



Black points show the ground measurements and red points show GEOS-Chem. Each gas is compared with carbon monoxide, and the horizontal axis represents chemical age. Steeper downward patterns indicate faster removal. Flatter patterns are consistent with continued formation partly offsetting chemical losses, although mixing may also contribute.

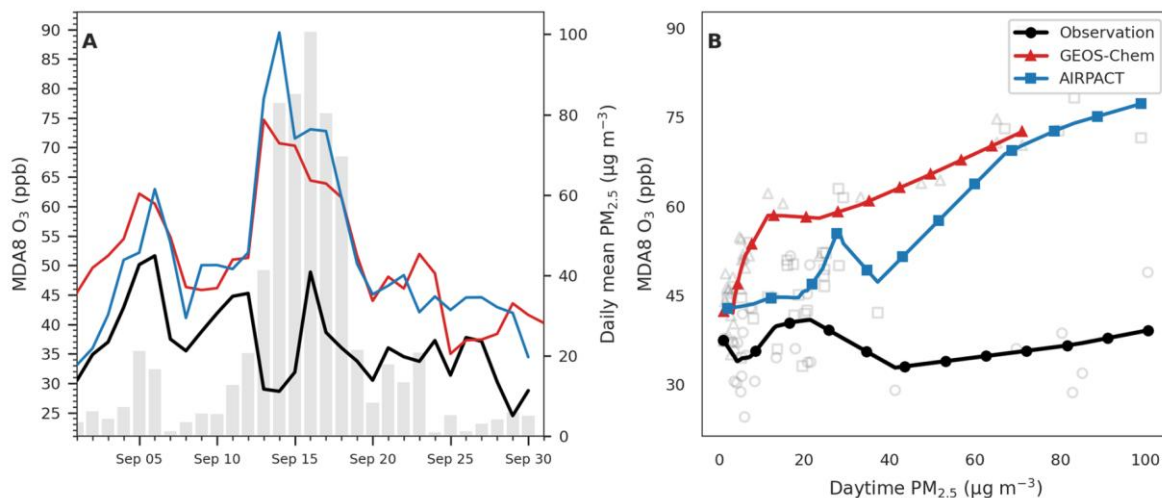
Adapted reading of Jin et al. (2026), Fig. 5, CC BY 4.0.

The smoke had changed along the way. The ozone measurements also showed a pattern reported in earlier wildfire studies: more smoke particles did not always mean more ozone.

OZONE IN AGED SMOKE

Why Did Ozone Stop Rising in the Heaviest Smoke?

Wildfires do not emit ozone directly. Smoke brings gases that can form ozone, but dense smoke can also block the sunlight that drives those reactions.



Black represents the Missoula observations, red represents GEOS-Chem, and blue represents AIRPACT. In panel B, ozone levels off and then declines as daytime PM_{2.5} increases, while both models continue upward.

Jin et al. (2026), Fig. 6, CC BY 4.0.

Earlier studies have shown that ozone can level off or decline in dense wildfire smoke. During the regional aged-smoke events in September 2020, our ground measurements in Missoula captured the same pattern. On lighter-smoke days, the day's highest eight-hour average ozone level increased with PM_{2.5}. In the heaviest smoke, it leveled off or fell. During 6–7 September, this ozone measure decreased by up to about 15 ppb while incoming sunlight fell by roughly 50%.

We could not identify a single cause. Weaker sunlight, lower temperatures, a shallower mixing layer, and no concurrent increase in nitrogen oxides may all have contributed. Both GEOS-Chem and AIRPACT predicted that ozone would continue rising as PM_{2.5} increased. GEOS-Chem reproduced the timing of the first two smoke events, but neither model captured the observed leveling or decline. In this case, getting the smoke timing right was not enough to reproduce its ozone response.

Caveat: The apparent change near 30–40 µg/m³ is based on a small number of smoke days. It should not be treated as a universal threshold for wildfire smoke.

Ozone was only one part of the smoke mixture. We next asked how its fine particles and hazardous gases shaped the health-risk estimates.

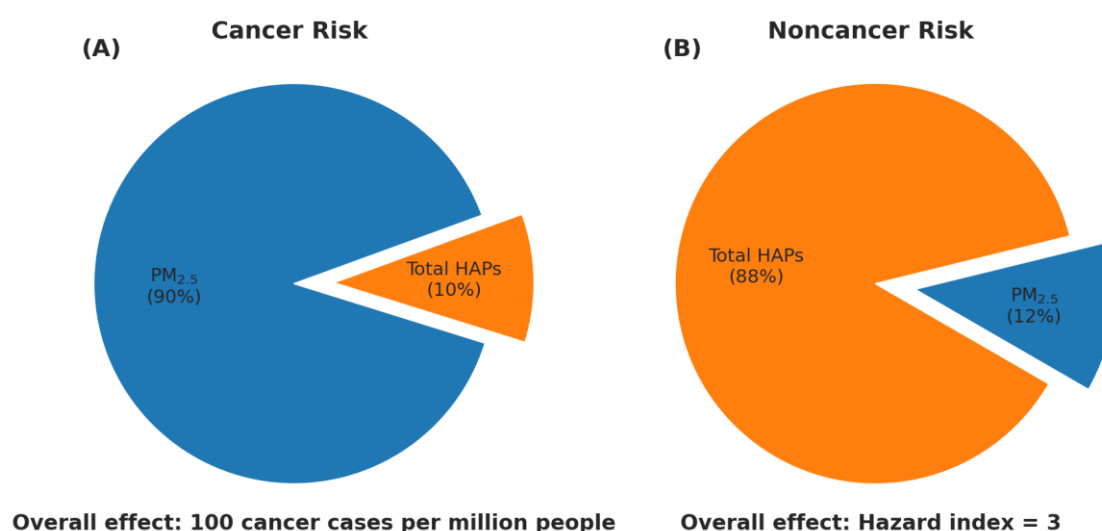
THE HEALTH PICTURE

PM_{2.5} Is Not the Whole Health Story

Fine particles drove most of the estimated cancer risk while measured hazardous gases drove most of the chronic non-cancer risk.

These are screening estimates, not illnesses observed during the 2020 event. To compare particles and gases, we used an upper-limit scenario in which a smoke season as intense as 2020 occurred every year for 70 years. Under this scenario, the estimated excess lifetime cancer risk was about 100 cases per million people. PM_{2.5} contributed about 90% of that estimate.

For chronic effects other than cancer, the pattern reversed. Measured hazardous air pollutants contributed about 90% of a hazard index of 3. Acrolein and formaldehyde were the largest contributors.



Fine particles contributed 90% of the estimated cancer risk, while measured hazardous gases contributed 88% of the chronic non-cancer hazard index.

Jin et al. (2026), Fig. 7, CC BY 4.0.

Caveat: The estimate of 100 cases per million represents lifetime risk under the repeated-season scenario. It is not the number of illnesses observed after the 2020 event. A hazard index of 3 does not mean that illness is three times more likely.

Because the two risk estimates depended on different pollutants, we next asked how well a widely used air-quality model reproduced both particles and gases.

TESTING GEOS-CHEM

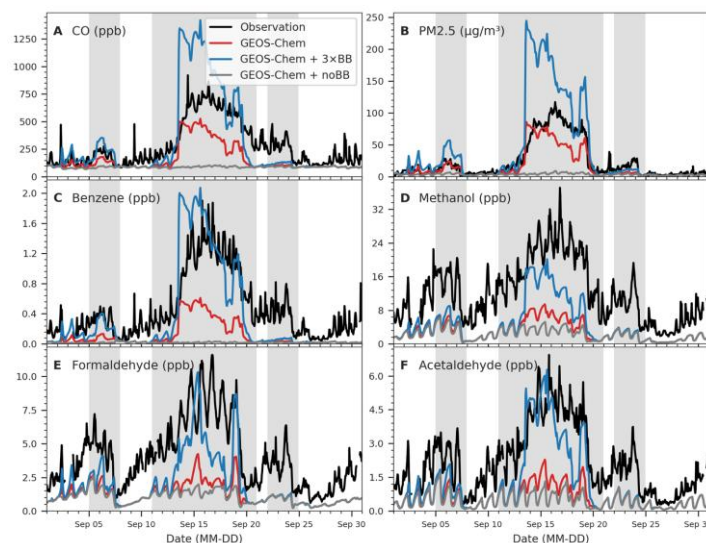
The Model Captured the First Two Smoke Events but Underestimated Pollution

GEOS-Chem captured when the first two smoke events arrived but underestimated many pollutants measured at ground level.

During the first two smoke events, GEOS-Chem closely followed the hour-to-hour timing of the observations. However, it did not reproduce the observed pollution enhancements during the third event. During smoke-affected periods, the model underestimated carbon monoxide and PM_{2.5} by about 30–40%, directly emitted aromatic gases such as benzene by 60–80%, and many oxygenated gases by 50–90%.

We tripled the modeled biomass-burning VOC and carbon monoxide emissions to test whether emissions alone explained these differences. This lowered the modeled OH and reduced the OH-exposure error from about 100% too high to about 30% too high. It brought some directly emitted gases closer to the observations, but shifted PM_{2.5} from about 30% too low to about 80% too high. Applying one uniform scaling to the modeled biomass-burning VOC and carbon monoxide emissions therefore could not correct the gases and particles at the same time.

These errors also carried into the screening-level health estimates. The base GEOS-Chem simulation yielded a cancer-risk estimate of 63 cases per million people, compared with 100 cases per million based on the observations. The estimated chronic hazard index was about 0.3 rather than 3.



Black represents the observations and red represents the base GEOS-Chem simulation. Blue shows the simulation with biomass-burning VOC and carbon monoxide emissions tripled, while gray shows the simulation with biomass-burning emissions turned off. The tripled-emissions test improved some gases but overestimated PM_{2.5} and did not correct every pollutant.

Jin et al. (2026), Fig. 8, CC BY 4.0.

In this case, getting the timing right was not enough. GEOS-Chem still underestimated ground-level pollution and screening-level health risks.

Explore the study: [ACP article](#) · [Supplement](#) · [Data](#) · [Analysis code](#)

This case study covers one city and one exceptional fire season. Its numerical estimates should not be transferred directly to every fire or location.